SELECTED SYLLABUS

Conservation Biology & Sustainable Development

ENV/BIO 3009/4900 | Natalia Borrego | Baruch College, CUNY | Spring 2018

> **About this copy.** This is a shortened version of the course syllabus I used at Baruch College. The course description, structure, assignments, grading, and representative activities are retained; contact information, dates, and institution-specific policy text have been removed for the website.

COURSE DESCRIPTION

Conservation biology is an interdisciplinary field concerned with understanding, protecting, and managing biological diversity. In this course we consider how biodiversity is measured, why it matters, the processes that threaten it, and the tools used to manage species, populations, ecosystems, and landscapes.

Because conservation and resource-management projects take place in human landscapes, ecological questions are considered alongside social, legal, and cultural context. The course combines classroom discussion with laboratory and field activities and uses several focal species and case studies to connect ideas across the semester.

COURSE OBJECTIVES

1. Describe the foundations of conservation biology and its relationship to ecological and human dimensions of environmental problems.
2. Describe and quantify biodiversity and explain how different measures capture different aspects of biological diversity.
3. Explain major processes contributing to biodiversity decline.
4. Evaluate conservation problems and potential responses across scales, from genes and populations to ecosystems and regions.
5. Use field, laboratory, and quantitative tools introduced in the course to investigate ecological questions.
6. Apply course concepts to conservation and management examples in New York and elsewhere.

COURSE STRUCTURE

Classroom

Short presentations are paired with discussion of readings and case studies. Class meetings extend rather than simply repeat assigned material.

Laboratory and field

Students work with existing datasets, collect information on biodiversity and conservation problems, use quantitative and spatial tools, and meet practitioners or visit sites relevant to course questions.

Quantitative work

Exercises include R, spreadsheet analysis, biodiversity metrics, population viability analysis, matrix models, and introductory spatial/GIS activities.

Open resources

The original course was part of CUNY's zero-textbook initiative. Required texts and additional readings were available freely or through the course site.

CORE RESOURCES USED IN THE COURSE

- *Conservation Biology for All*, eds. N. S. Sodhi and P. R. Ehrlich.
- *Spreadsheet Exercises in Conservation Biology and Landscape Ecology*, T. M. Donovan and C. W. Welden.
- Primary literature, case-study modules, podcasts, and other open instructional resources assigned throughout the semester.

ASSIGNMENTS AND ASSESSMENT

Students encounter conservation questions through discussion, data, field work, and scientific writing.

COURSE ASSIGNMENTS

COMPONENT	DESCRIPTION
Homework and classwork	Reading quizzes, short written responses, case-study discussion, laboratory activities, and other preparation or in-class work.
Exams	Two examinations covering assigned readings and material developed in class. The final was cumulative in the original course.
Writing assignment	A written assignment submitted electronically; details were provided through the course site.
Research project option	Students enrolled in the advanced BIO/ENV 4900 option worked in teams to collect and analyze data related to a current conservation topic, gave a short presentation, and submitted a group paper. The research-project option was also open to interested students enrolled in BIO/ENV 3009.

ORIGINAL GRADING STRUCTURE

ENV/BIO 3009

Exam I	25%
Final exam	35%
Homework and classwork	30%
Writing assignment	10%

BIO/ENV 4900 / research option

Exam I	20%
Final exam	30%
Homework and classwork	30%
Writing assignment	10%
Group project	10%

EXAMPLES OF COURSE ACTIVITIES

Biodiversity and data

Biodiversity measurement; introductory R work; spreadsheet exercises; interpreting patterns across scales.

Human dimensions

Environmental justice, ecological economics, and discussion with an environmental lawyer.

Spatial ecology

Introductory GIS / Google Earth, ecosystem fragmentation, landscape-scale conservation questions.

Population tools

Conservation genetics, population viability analysis, applied demography, and matrix models.

Field and professional context

Zoo and Harbor School visits and a United Nations trip were included in the 2018 schedule.

Student communication

Student presentations on emerging threats, conservation case studies, and group research results.

REPRESENTATIVE COURSE SEQUENCE

Questions and tools build across biological scales.

COURSE UNIT	TOPIC	REPRESENTATIVE ACTIVITIES / MATERIALS
Foundations	**History of conservation ecology; what biodiversity is and why it matters**	Scientific-paper reading; introductory R; biodiversity modules and metrics.
Human dimensions	**Conservation in social, economic, legal, and cultural context**	Environmental justice; ecological economics; practitioner discussion.
Habitat change	**Fragmentation and landscape structure**	Introductory GIS / Google Earth; ecosystem-fragmentation activities.
Exploitation & invasion	**Harvest, invasive species, and population processes**	Overexploitation exercises; applied demography; harvested-population management.
Climate & genetics	**Novel threats and conservation at the genetic level**	Conservation-genetics activity; student presentations on emerging threats.
Population & species	**Population, species, and landscape management**	Population viability analysis; matrix models.
Protection & restoration	**Protected areas, endangered species, reintroductions, restoration**	Global reintroduction case studies; marine spatial planning.
Synthesis	**Sustainable use and applying course ideas to management decisions**	Resource-use questions; group research presentations; course synthesis.

WHAT I WOULD RETAIN IN A CURRENT VERSION

The specific readings and software would be updated, but I would retain the course's basic structure: students move among field observation, quantitative analysis, primary literature, and discussion rather than encountering conservation as a list of threats and solutions. Local field examples would change with the institution; the central questions and analytical skills would remain.